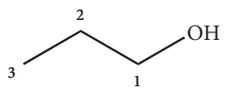
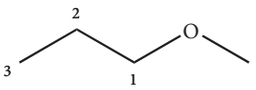
4.

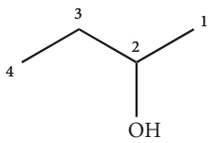
$C_4H_{10}O$ isomers



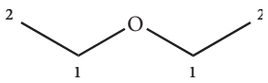
propan-1-ol



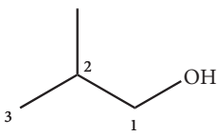
1-methoxypropane



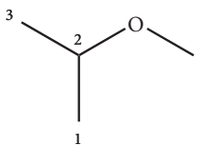
butan-2-ol



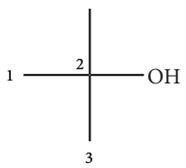
ethoxyethane



2-methylpropan-1-ol



2-methoxypropane



2-methylpropan-2-ol

5. Weight of organic substance w = 0.2346 g

Weight of water (x) = 0.2754 g

Weight of CO_2 (y) = 0.4488 g

$$\text{Percentage of carbon} = \frac{12}{44} \times \frac{y}{w} \times 100$$
$$= \frac{12}{44} \times \frac{0.4488}{0.2346} \times 100$$
$$= 52.17\,\%$$

$$\text{Percentage of hydrogen} = \frac{2}{18} \times \frac{x}{w} \times 100$$
$$= \frac{2}{18} \times \frac{0.2754}{0.2346} \times 100$$
$$= 13.04\,\%$$

$$\text{Percentage of oxygen} = [100 - (52.17 + 13.04)]$$
$$= 100 - 65.21 = 34.79\,\%$$

6. Weight of organic substance (w)=0.16 g

Weight of Barium sulphate (x) = 0.35 g

$$\text{Percentage of Sulphur} = \frac{32}{233} \times \frac{x}{w} \times 100$$
$$= \frac{32}{233} \times \frac{0.35}{0.16} \times 100$$
$$= 30.04\,\%$$

7. Weight of organic substance (w)=0.185 g

Weight of silver bromide (x) = 0.320 g

$$\text{Percentage of bromine} = \frac{80}{188} \times \frac{x}{w} \times 100$$
$$= \frac{80}{188} \times \frac{0.32}{0.185} \times 100$$
$$= 73.6\,\%$$

Weight of organic substance (w)=0.40 g

Weight of silver iodide (x) = 0.235 g

$$\text{Percentage of iodine} = \frac{127}{235} \times \frac{x}{w} \times 100$$
$$= \frac{127}{235} \times \frac{0.235}{0.40} \times 100$$
$$= 31.75\,\%$$

8. Weight of organic substance (w)=0.33 g

Weight of $Mg_2P_2O_7$ (x) = 0.397 g

$$\text{Percentage of phosphorous} = \frac{62}{222} \times \frac{x}{w} \times 100$$
$$= \frac{62}{222} \times \frac{0.397}{0.33} \times 100$$
$$= 33.59\,\%$$

9. Weight of organic compound (w) = 0.3 g

11th Chemistry Key Answers Vol-2 EM.indd 306 09-01-2023 14:45:52